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LXS Demura Simulator User Guide

LX89507 (Cassandra-1)

2026. 03. 16

Laboratory
Mid/Small Display Image Quality IP Team

History

No.	Date	Description	Author
1	2026.01.30	Initial. • The usage is the same as the LX89504 Simulator, but the internal code has been updated.(page4) • In particular, the binary generation for LX89507 and the Corner-Cut Functions have been modified.	LXS
2	2026.03.16	• The Corner-Cut functions have been updated, including subdividing *CORNER_CUT_TOP* into *CORNER_CUT_TOP_H* and *CORNER_CUT_TOP_V.* • The input parameter file name has been changed from *lxs_para_sim_LX89507.txt* to *lxs_para_sim.txt*	LXS

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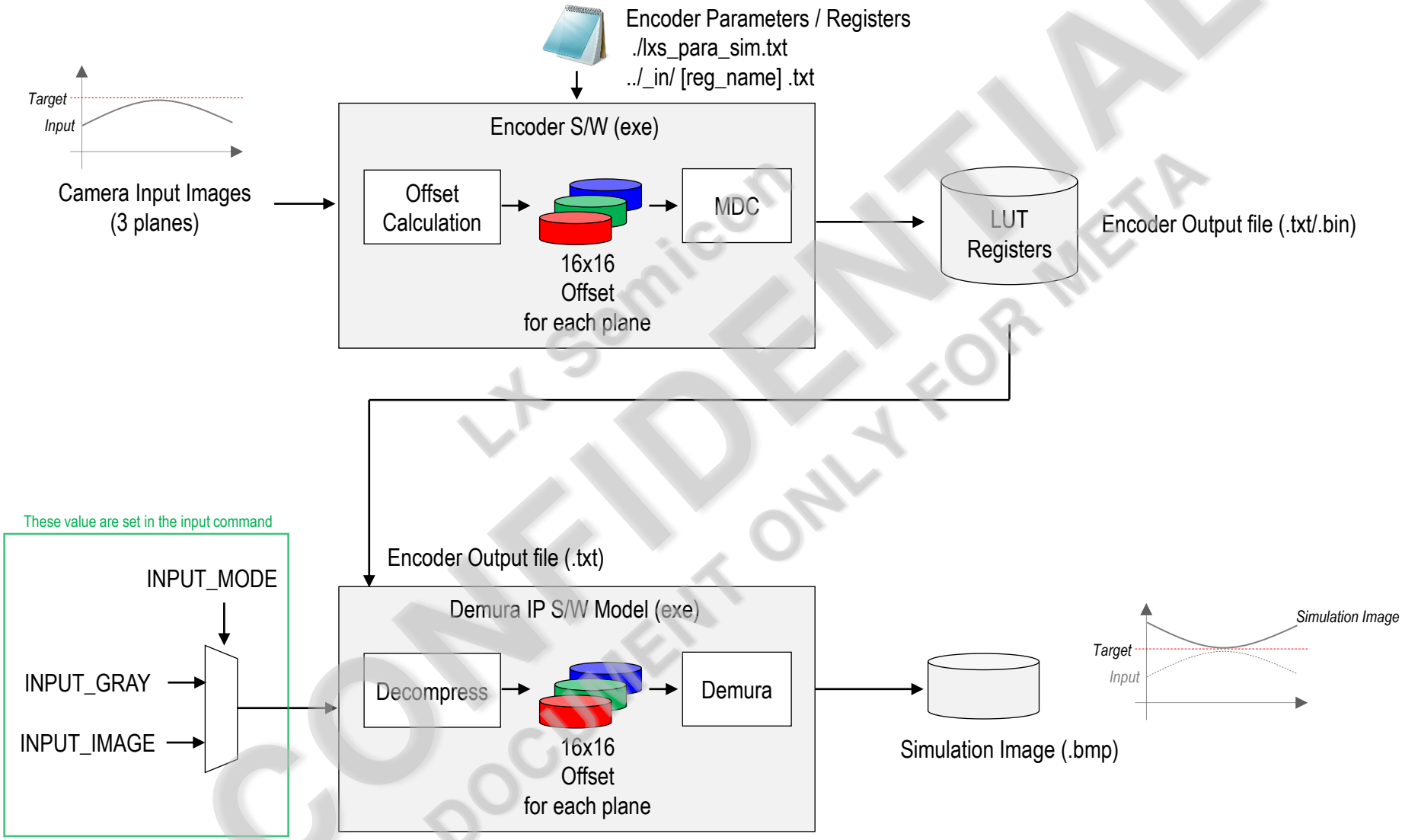
LXS Demura Simulator Flow



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Demura Simulator Interface Guide

Program Folder Hierarchy

```
LX89507_Demura_MDC_Simulator
├── Simulator
│   ├── LX89507_Demura_MDC_enc.exe
│   ├── LX89507_Demura_MDC_dec.exe
│   └── lxs_para_sim.txt
└── _in
    ├── camera_image
    │   ├── [panel_name] / [panel_name]_[{RED/GRN/BLU}][Gray]_DISP_RAW.png          IMG_BLOCK==0
    │   └── [panel_name] / [panel_name]_[{RED/GRN/BLU}][Gray]_DISP_RAW_BLOCK.png    IMG_BLOCK==1
    ├── register                                                                     (lxs_para_sim.txt)
    │   └── [reg_name] .txt
    └── display_image
        └── [display_image_name] . [format]
└── _out
    ├── lut
    │   ├── [lut_name].txt
    │   └── 1x1_[lut_name].txt   → If there is no 1x1 lut, the 1x1 output and PSNR will not be created
    ├── reg
    │   └── [reg_name] .txt
    └── image
        ├── [lut_name]_[reg_name]_[display_image_name]_comp.bmp
        ├── [lut_name]_[reg_name]_[display_image_name]_comp_1x1.bmp
        └── [lut_name]_[reg_name]_PSNR.txt
```

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Demura Simulator Command

- Demura Encoder

- LX89507_Demura_MDC_enc.exe [panel_name] [register_name]

- ex) LX89507_Demura_MDC_enc.exe C1,C3 reg_0,reg_1

- Demura Decoder

- [Input Mode0] Input == Solid Image
- [Input Mode1] Input == File Display Image

- LX89507_Demura_MDC_dec.exe [lut_name] [dec_register_name] [Input mode] [Input gray or display image]

- ex) LX89507_Demura_MDC_dec.exe C1,C3 reg_0,reg_1 0 64

- ex) LX89507_Demura_MDC_dec.exe . C1,C3 reg_0,reg_1 1 Disp_W_GRAYINC_8pix.bmp

../out/lut/[lut_name].txt
../out/lut/1x1_[lut_name].txt

../out/reg/[dec_register_name].txt

→ If there is no 1x1 lut, the 1x1 output and PSNR will not be created

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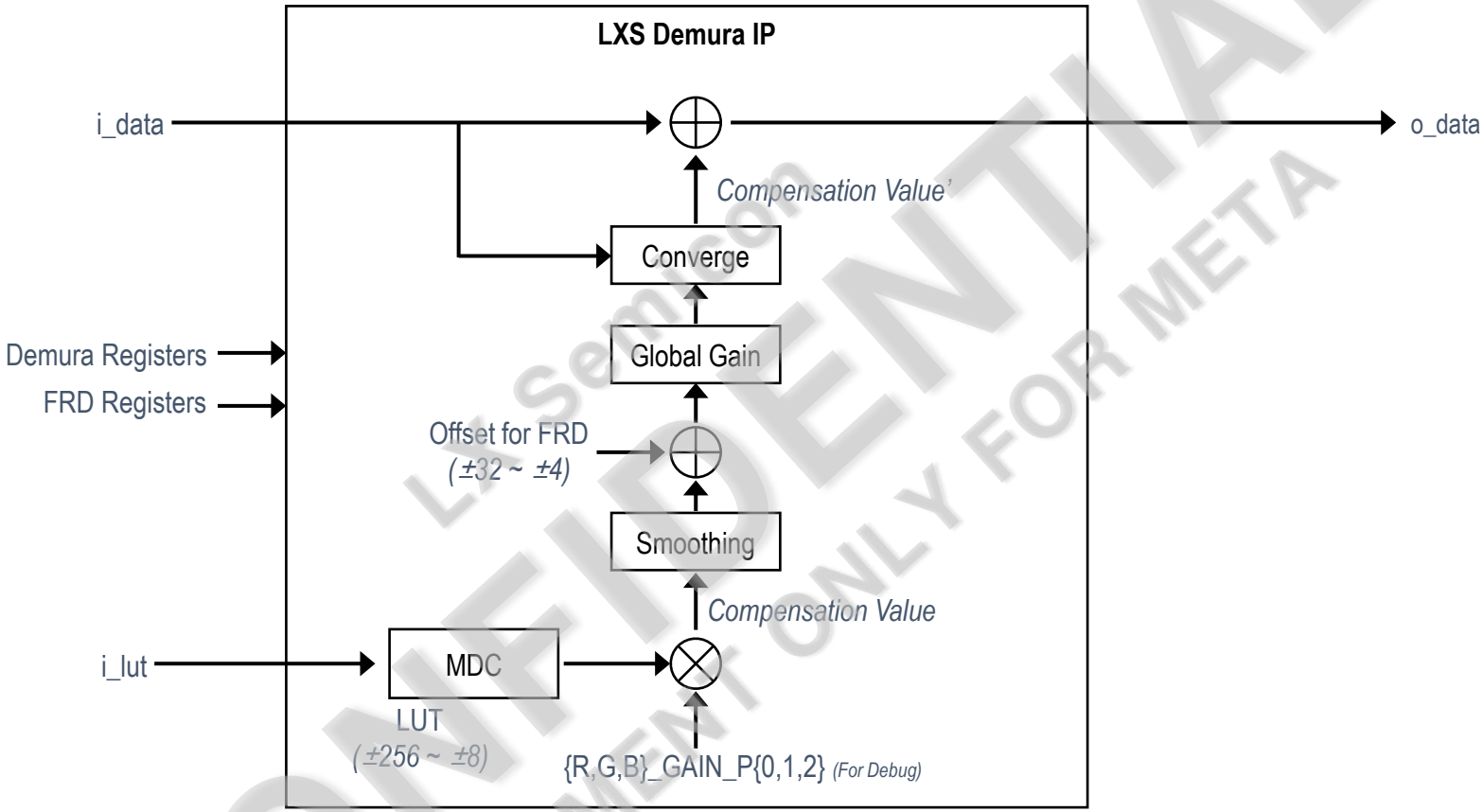
Demura IP Datapath



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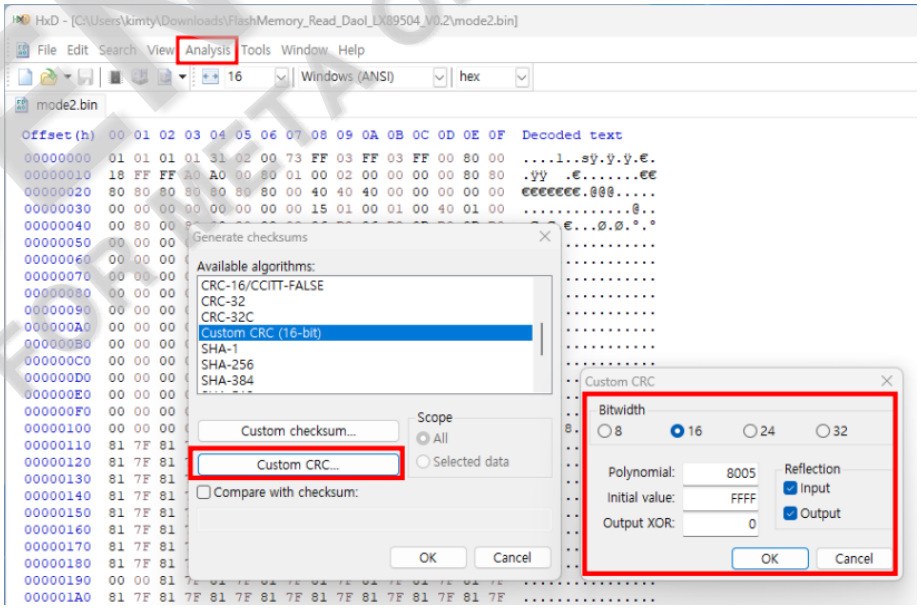
Binary File Configuration

- **Flash Size**
  - Header Size : 4 Byte
  - Demura CFG Size : 256 Byte
  - CRC for Header and CFG : 2 Byte
  - LUT data Size : Variable
  - CRC for LUT : 2 Byte

- **CRC (Cyclic Redundancy Check)**
  - CRC-16 Modbus
    - Polynomial : 0xA001
    - Initial Value: 0xFFFF
    - Reflection : Both the input data and the output CRC value
    - Bit Width : 16

  - How to check CRC
    1) Hxd Program
    2) Menu -> Analysis -> Generate checksums->Custom CRC
    3) Set Polynomial, Initial Value, Reflection, Bit Width



How to check CRC

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Binary File Configuration

- ## Flash Binary File (.bin)

Resolution 2328x2488

Address	Data
23'h00000	Header (4 Bytes)
23'h00004	Demura CFG Data (256 Bytes)
23'h00104	CRC for header & CFG (2 Bytes)
23'h00106	Demura LUT Data (73,300 Bytes)
23'h11E52	CRC for Demura LUT (2 Bytes)
23'h11E54	Dummy

Demura Mode0
(RGB Mode, Plane 2, Compression)

Address	Data
23'h00000	Header (4 Bytes)
23'h00004	Demura CFG Data (256 Bytes)
23'h00104	CRC for header & CFG (2 Bytes)
23'h00106	Demura LUT Data (77,056 Bytes)
23'h12CFE	CRC for Demura LUT (2 Bytes)
23'h12D00	Dummy

Demura Mode1
(RGB Mode, Plane 3, Compression)

Address	Data
23'h00000	Header (4 Bytes)
23'h00004	Demura CFG Data (256 Bytes)
23'h00104	CRC for header & CFG (2 Bytes)
23'h00106	Demura LUT Data (73,744 Bytes)
23'h12008	CRC for Demura LUT (2 Bytes)
23'h12010	Dummy

Demura Mode2
(White Mode, Plane 3, No Compression)

- ## Example

```
Offset(h)  00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

00000000   01 01 01 01 31 12 02 00 FF 03 FF 03 FF 00 00 00
00000010   00 FF FF A0 A0 00 C0 01 7A 02 80 00 00 00 80 80
     ⋮
00000100   00 00 00 00 93 C6 81 7F 81 7F 81 7F 81 7F 81 7F
00000110   81 7F 81 7F 81 7F 81 7F 81 7F 81 7F 81 7F 81 7F
     ⋮
0000F750   7F 81 7F 81 7F 81 7F 81 7F 81 00 00 00 00 00 00
0000F760   00 00 7F 81 7F 81 CD C2
```

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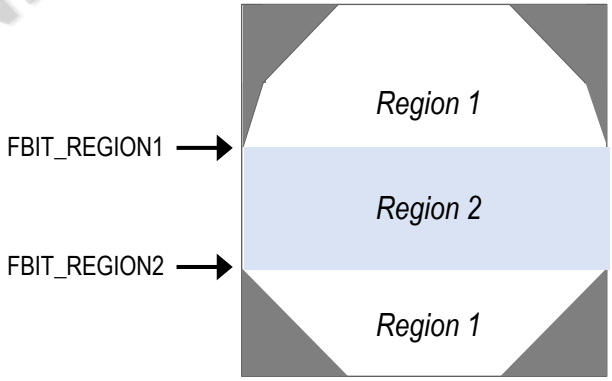
FBIT

- ## LUT Fraction Bit (FBIT)
  - LUT Bit : 8bit
  - Compensation Range : ±256 ~ ±8

PLANE{00~02}_FBIT_{R1, R2}	Range	Precision	Note.
0x4	-8.9375 ~ +7.9375	0.0625	Q(4. 4), 4s4
0x3	-16.875 ~ +15.875	0.125	Q(5. 3), 5s3
0x2	-32.75 ~ +31.75	0.25	Q(6. 2), 6s2
0x1	-64.5 ~ +63.5	0.5	Q(7. 1), 7s1
0x0	-128.0 ~ +127.0	1	Q(8. 0), 8s0
0xF	-256.0 ~ +254.0	2	Q(9. -1), 9s-1

- ## LUT Fraction Bit Region
  - Set FBIT_REGION1, FBIT_ REGION2 registers, unit is 16pixels.
  - Compensate ROI*(Region Of Interest)*, more precisely.
  - Smoothing off → Available FBIT_R1, R2
  - Smoothing on → Only FBIT_R1



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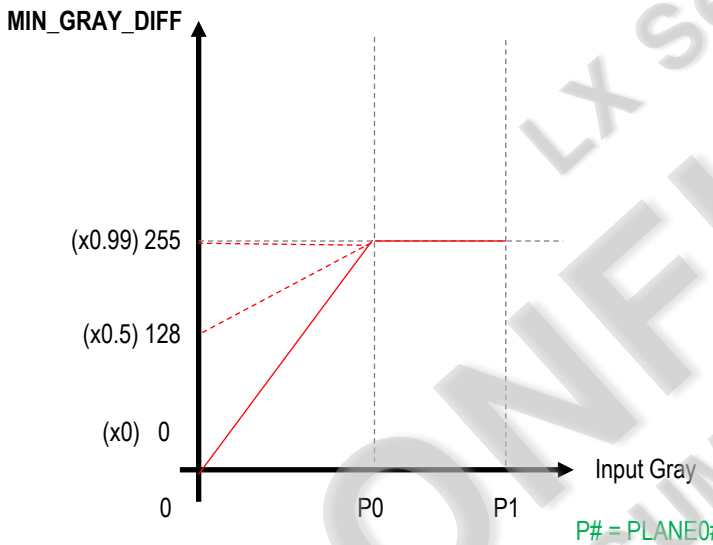
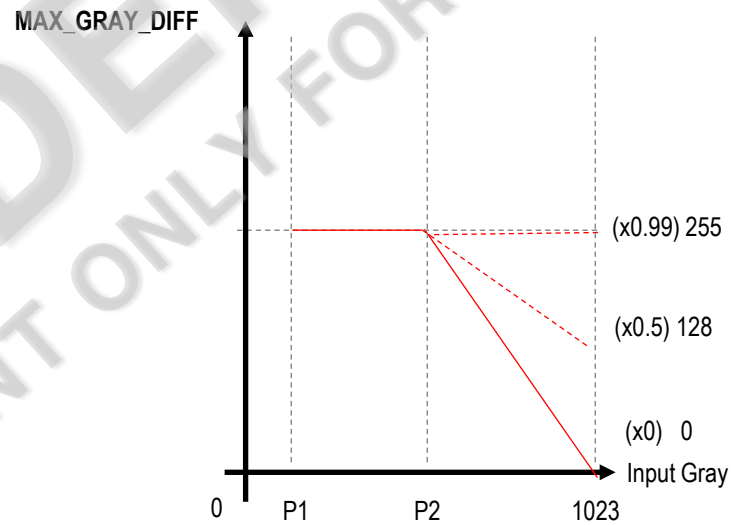
Extrapolation Function

- **Extrapolation**
  - MIN_GRAY_DIFF
    - Adjust the weights of low gray precisely by controlling the slope of the compensation strength below Plane0
    - 8bit, x0~ x0.99 scale (x0.99 = 255).
  - MAX_GRAY_DIFF
    - Adjust the weights of low gray precisely by controlling the slope of the compensation strength above Plane2
    - 8bit, x0~ x0.99 scale (x0.99 = 255).





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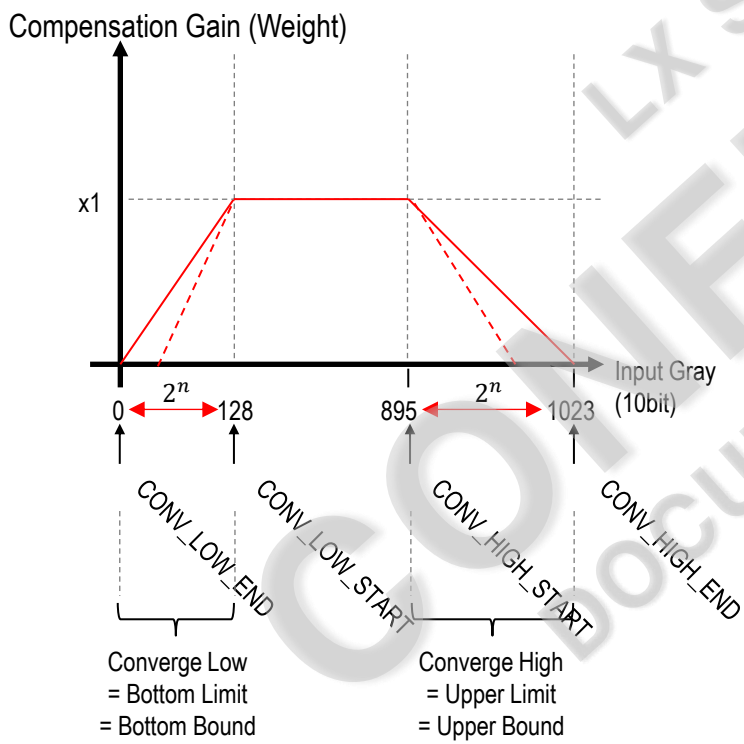
Converge Function

- **Convergence**
  - Adjust the weights to make the Compensation Data converge to 0 in the Low Gray and High Gray regions
  - The register related to converge should be set as follows:
    - 0 ≤ CONV_LOW_END ≤ CONV_LOW_START < CONV_HIGH_START ≤ CONV_HIGH_END ≤ 1023
    - CONV_HIGH_END – CONV_HIGH_START should be power of 2.
    - CONV_LOW_START – CONV_LOW_END should be power of 2.
  - If you do not want to use converge, you can change it through the following the register of converge enable
    - CONV_LOW_EN, CONV_HIGH_EN
  - Using Shift Interpolation, G/C Decrease.
    - 320K → 81K



$$Compensation\ Value' = Compensation\ Value\ * Gain(Weight)$$

CONV_LOW_END	CONV_LOW_START	CONV_HIGH_START	CONV_HIGH_END
0~1023	0~1023	0~1023	0~1023

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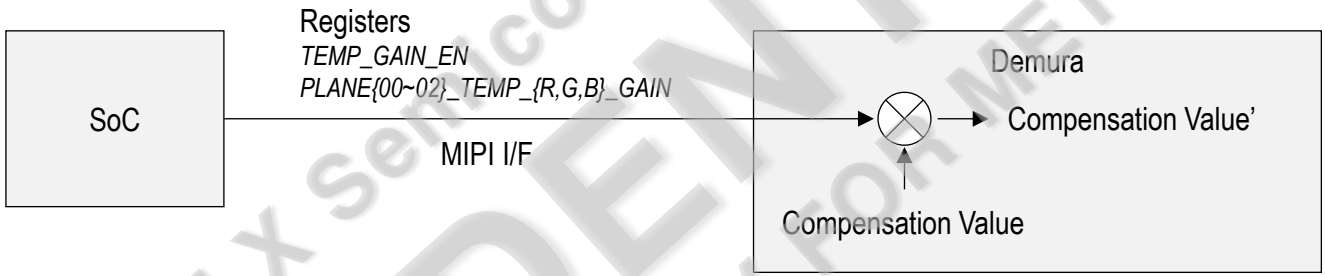
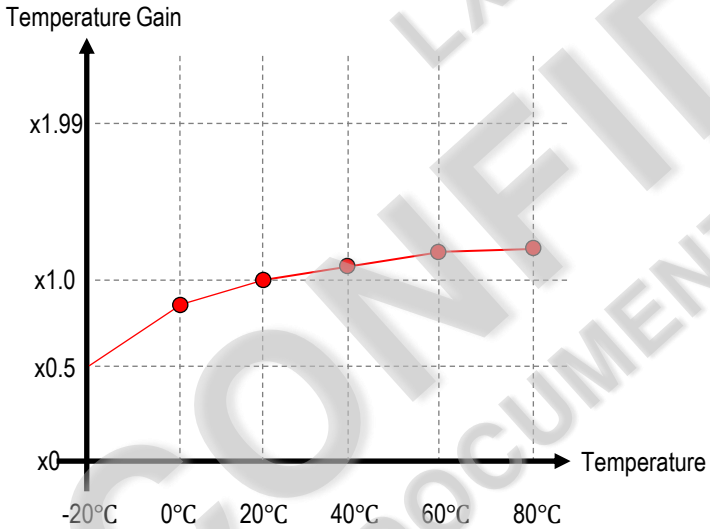
Temperature Gain Function

- **Temperature Gain (Global Gain)**
  - Compensation Strength Control for Temperature
  - Temperature Gain, 8bit
  - TEMP_GAIN_EN
  - PLANE{00~02}_TEMP_{R,G,B}_ GAIN, 8bit
  - x0.0 ~ x1.9921875





<Temperature Gain Graph (Example)>

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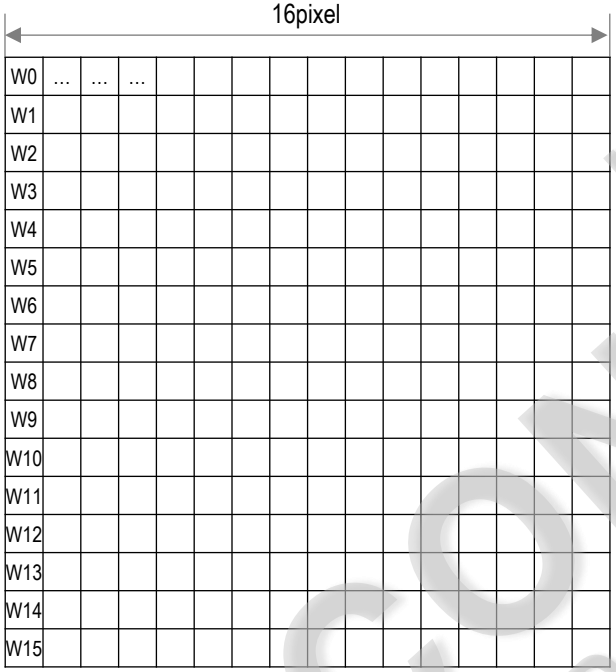
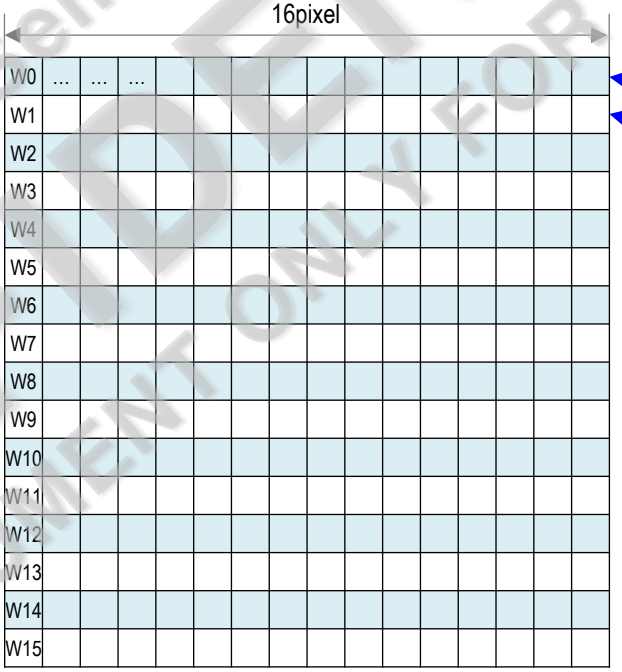
Smoothing Function with FRD

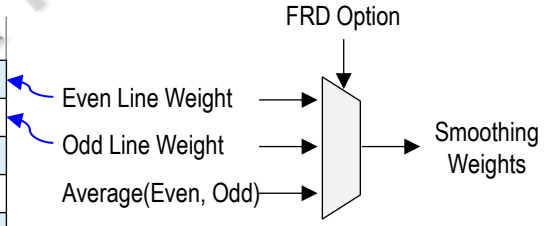
- **FRD Option**
  - The register for how to apply LUT to the GGS2 area in FRD On
  - FRD_OPT
    - 0x00 : Even Line Weight
    - 0x01 :Odd Line Weight
    - 0x02 : Average (Even, Odd Line)

Smoothing On + FRD Off (GGS2 Area)



Smoothing On + FRD On (GGS2 Area)





∵ *Differences on physical position.*

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FRD Offset

- **FRD Offset**
  - FRD_OFST_{R,G,B}_P{0~2}
  - 8bit

- **FRD Offset Fraction Bit**

FRD_OFST_P{0~2}_{R,G,B}	Range	Precision	Note.
0x3	-4.96875 ~ +3.96875	0.03125	Q(3. 5), 3s5
0x2	-8.9375 ~ +7.9375	0.0625	Q(4. 4), 4s4
0x1	-16.875 ~ +15.875	0.125	Q(5. 3), 5s3
0x0	-32.75 ~ +31.75	0.25	Q(6. 2), 6s2

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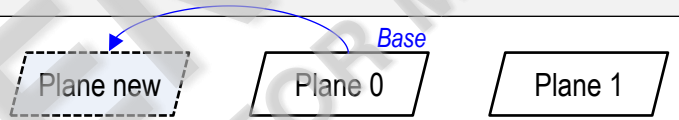
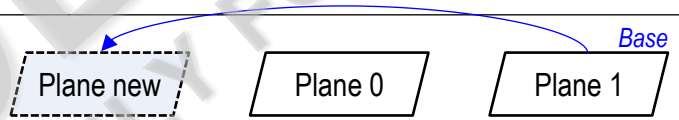
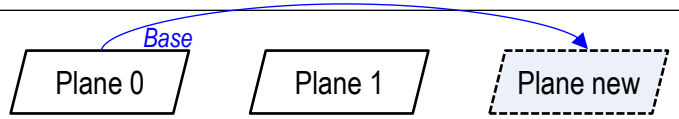
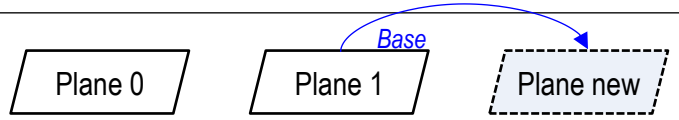
Offset Approximation Support

- **Make LUT from other LUT of base plane when DMR_MODE==1.**
- **Register** (plane_num ==2)
  - APRM_TARGET_PLANE : New plane that want to make
  - APRM_BASE_PLANE : Reference plane
  - APRM_GAIN_R : Gain for red (x0 ~ x3.984375)
  - APRM_GAIN_G : Gain for green (x0 ~ x3.984375)
  - APRM_GAIN_B : Gain for blue (x0 ~ x3.984375)

Case	APRM_ TARGET_PLANE	APRM_ BASE_PLANE	Operation
1	0	0	
2	0	1	
3	1	0	
4	1	1	
5	2	0	
6	2	1	

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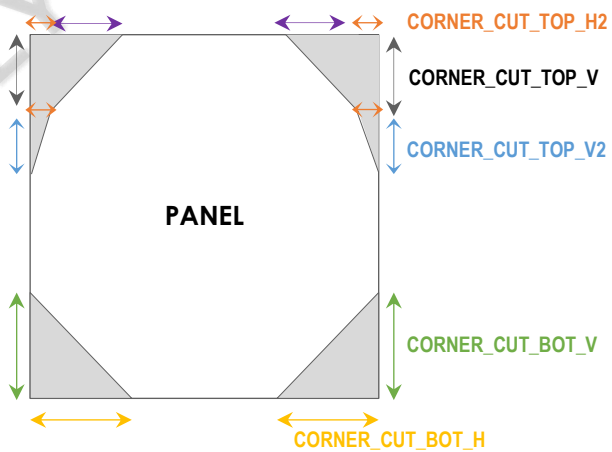
lxs_para_sim.txt

Parameter	Default	Min	Max	Description
BLOCK_MEDIAN	1	0	1	Method for converting 1x1 data to block size x block size 0: Average of the data 1: Median of the data
TARGET_RADIUS	111	1	-	To determine the target value for Demura, a circular area is selected at the center of the panel. The average value of the pixels within the circle is used as the target value, and the radius of the circle is also determined as TARGET_RADIUS.
CORNER_CUT_TOP_H	420	0	-	The distance between a vertex of the diagram that represents the panel and the corner of the rectangle that encloses the panel. Please refer to the image below. 
CORNER_CUT_TOP_V	420	0	-	
CORNER_CUT_TOP_H2	145	0		
CORNER_CUT_TOP_V2	342	0	-	
CORNER_CUT_BOT_H	487	0	-	
CORNER_CUT_BOT_V	489	0	-	

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lxs_para_sim.txt

Parameter	Default	Min	Max	Description
BLOCK_SIZE	16	1	-	Block size (The Demura IP only supports BLOCK_SIZE==16)
IMG_BLOCK	1	0	1	Input image selection mode 0: Full resolution Image ([panel_name] / [panel_name]_[Color][Gray]_DISP_RAW.png) 1: 16x16 blocked Image ([panel_name] / [panel_name]_[Color][Gray]_DISP_RAW_BLOCK.png)
ANALYSIS_MODE	0	0	1	Intermediate data is saved for analysis. 0 : nothing happened 1 : data is saved in '../_out/plot/' path
INPUT_PNG	1	0	1	The format of the input camera image. 0 : '**csv**' format 1 : '**png**' format
OUTPUT_TXT	1	0	1	The format of the output LUT. 0 : The LUT and registers are saved in a '**.bin**' file. 1 : The LUT and registers are saved in separate '**.txt**' files.
FBIT_AUTO	1	0	1	Fraction bit Setting method – Selection mode 0 : Manually fix the FBIT registers using the value in "LXS_reg.ini" 1 : Automatically set up the FBIT registers in the encoder.

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